2.2 Proteins

Proteins are the fundamental biological components of skin, hair, muscles, connective tissues, enzymes etc...

2.2.1 Definition

Proteins are defined as the biopolymers composed of α-amino acids linked by peptide bonds. They are also known as polypeptides. Proteins are the major constituents of all living organisms. They contain carbon, hydrogen, nitrogen, oxygen and sulphur.

2.2.2 Classification

Proteins are classified into three main groups.

a. Simple protein:

The protein that yields only α-amino acids on hydrolysis is called a simple protein. Examples: albumin and globulin

b. Conjugated protein:

The protein that yields α-amino acids and a non-protein part upon hydrolysis is called a conjugated protein. The non-protein part is called as the prosthetic group.

Based on the nature of prosthetic group, conjugated proteins are further classified as follows.

i Nucleoprotein:

Protein associated with nucleic acids is called as nucleo protein.

Example: Histone(in DNA).

ii. Phosphoprotein:

Protein containing phosphoric acid is called as phosphoprotein.

Example: casein (in milk).

iii. Glycoprotein:

Protein containing carbohydrate moiety is called as glycoprotein.

Example: mucin (in saliva)

Unit 2.indd 60 21-05-2018 15:08:13